

Lesson 3 Lie's thm & Jordan-Chevalley decomposition

Wednesday, March 4, 2026 20:58

Thm Lie's thm

the restriction
→ is from lemma

$\text{Char } F = 0$ or $\text{Char } F > \dim_F M$.

Let L be a solvable Lie algebra and M a module over L , $\dim_F M = n < \infty$

Then $\exists$ a basis of M s.t. $L \subseteq \begin{pmatrix} * & * \\ 0 & * \end{pmatrix}$ Let M_1 be a minimal submodule $\Rightarrow M_1 = Fm_1$. Consider M/Fm_1 . Let $M_2 \subseteq M/Fm_1$ be a minimal submodule. $Lm_1 \subseteq Fm_1 + Fm_2$

($\Leftrightarrow$) any irreducible f.d. L -module has dimension 1.)

pf: Let M be an irreducible f.d. L -module.

As we are mainly concerned with the image of $L \rightarrow \text{End}_F(M)^{(-)}$, we may assume $L \subseteq \text{End}_F(M)^{(-)}$.

Since L is solvable, we have

$$L > L^{[1]} = [L, L] > L^{[2]} > \dots > L^{[s]} > L^{[s+1]} = (0) \text{ for some } s.$$

If $s=0 \Rightarrow L$ is abelian $\Rightarrow \exists$ an eigenvector v for all $x \in L$

If $s \geq 1$, pick any $a \in L^{[s]}$, $b \in L$, then $[a, [a, b]] \in L^{[s+1]} = (0) \Rightarrow [a, b]^n = 0$.

Consider $S = \{[a, b] \mid a \in L^{[s]}, b \in L\}$. This is a Lie subset

By Engel's thm, $\text{Span}_F(S) = [L^{[s]}, L]$ is nilpotent.

Note that for any ideal $I \in L$, IM is a submodule, hence $IM = (0)$ or M .

Thus we must have $[L^{[s]}, L] = (0)$, i.e. $L^{[s]}$ lies in the center of L .

Now, for any $a \in L^{[s-1]}$, $b \in L^{[s-1]}$, $[a, [a, b]] = 0 \Rightarrow [a, b]$ is nilpotent.

Apply again Engel's thm, we conclude that $L^{[s]} = (0)$, a contradiction.

Rmk a third version

L solvable, $\text{Char } F = 0$ or $> \dim_F M$.

(1) $\forall$ irreducible f.d. L -module is 1-dim.

(2) If M is f.d. L -module, then $\exists$ a basis in M s.t. $L \subseteq \begin{pmatrix} * & * \\ 0 & * \end{pmatrix}$

(3) If M is any L -module, then $[L, L]$ acts nilpotently

$$(2) \Rightarrow (3) \quad \left[\begin{pmatrix} * & * \\ 0 & * \end{pmatrix}, \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \right] \subseteq \begin{pmatrix} 0 & * \\ 0 & 0 \end{pmatrix}$$

(3) $\Rightarrow$ (1) $[L, L]M = (0) \Rightarrow [L, L] = 0 \Rightarrow \exists$ a common eigenvector.

Example

$\text{Char } F = 2$ $\mathfrak{gl}(2, F)$ is solvable

(commutation is of the form

$$\begin{pmatrix} a & x \\ 0 & y \end{pmatrix} \begin{pmatrix} a & x \\ 0 & y \end{pmatrix} \Rightarrow [^2] \subseteq \begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix} \Rightarrow [^3] = (0)$$

(commutation is of the form

$$\begin{pmatrix} \alpha & \gamma \\ \beta & -\alpha \end{pmatrix} = \begin{pmatrix} \alpha & \gamma \\ \beta & \alpha \end{pmatrix} \Rightarrow L^{[2]} \subseteq \left\{ \begin{pmatrix} \alpha & 0 \\ 0 & \alpha \end{pmatrix} \right\} \Rightarrow L^{[3]} = (0)$$

Jordan - Chevalley decomposition

let V be any f.d. vector space, $\varphi \in \text{End}_F(V)$

let $\alpha_1, \dots, \alpha_s$ be all distinct eigenvalues of φ

① $V = V_{\alpha_1} \oplus \dots \oplus V_{\alpha_s}$ where V_{α_i} is the root space w.r.t. α_i .

let $\chi(t)$ be the characteristic polynomial of φ

$$= (t - \alpha_1)^{n_1} \dots (t - \alpha_s)^{n_s}$$

let $P_i(t) = \frac{\chi(t)}{(t - \alpha_i)^{n_i}}, 1 \leq s$

Then $\exists f_1, \dots, f_s$ s.t. $\sum_{i=1}^s P_i(t) f_i(t) = 1$

$$\forall v \in V, \quad v = \underbrace{p_1(\varphi) f_1(\varphi)(v)}_{\in \ker(\varphi - \alpha_1 \text{Id})^{n_1}} + \dots + p_s(\varphi) f_s(\varphi)(v)$$

(in fact, we can use any annihilating polynomial; existence guaranteed by V f.d.)